

Utilizing Graph Neural Networks (GNN) in Quantum-Natural Language Processing (Q-NLP) for Risk Management in Banking Sector: A Novel Approach

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Abstract

The field of Natural Language Processing (NLP) has grown in large numbers since the 1940s, beginning with early machine translation. Despite the progress that has been seen in this field, there are several challenges that still exist in understanding and generating human language. This chapter presents the historical evolution of NLP and presents a summary of all its primary elements, which include Natural Language Generation (NLG) and Understanding (NLU), as well as its subfields, which include syntactic, morphological, and semantics. We also look at the several uses of NLP that are present in conversational systems. Such systems allow for automated customer interactions and provide support that is available 24/7; machine translation, which offers a real-time communication channel in different languages for international banking; information extraction, which helps in analyzing large volumes of financial information and documents; and named entity recognition which is extremely valuable when it comes to identifying key entities such as companies, individuals, and financial instruments in regulatory reports. Additionally, we introduce the application of Graph Neural Networks (GNNs) in NLP, highlighting their use within